QP CODE: 25900035

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

SECOND SEMESTER MGU-UGP (HONOURS)

REGULAR EXAMINATION APRIL 2025

Second Semester

Discipline Specific Core Course–MG2DSCCND100

FUNDAMENTALS OF FOOD SCIENCE

(2024 ADMISSION ONWARDS)

Duration: 2 Hours Maximum Marks: 50

Remember(K),Understand(U),Apply(A),Analyse(An),Evaluate(E),Create(C),Interest (I),Appreciation(Ap),and Skill(S)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Part A
Multiple ChoiceQuestions
Answer all questions
Each question carries 1 mark

1. Chemical ***(1 mark)***
2.Aerosol ***(1 mark)***
3. Melting ***(1 mark)***
4. Smoke Point ***(1 mark)***
5. Recipe yield ***(1 mark)***
6. Removal of moisture ***(1 mark)***

Part B
Short Answer Type Questions.
Answer any twoquestions.
Each question carries 2 marks.

7. A recipe which is tested and gives the same product every time when it is tried under the same circumstances. (Any definition) ***(2 marks)***

8.

- Washing cereals causes loss of B complex vitamins- ***(1 mark)***
- Water soluble vitamins are lost when fruits and vegetables arecut and then washed-***(1 mark)***

9.

- Food science ensures a complete understanding of food safety standards and helps to identify the risk of contamination- ***(1 mark)***
- Food science makes it possible to maximize our food resources and minimize food wastage-***(1 mark)***

10.

- Food does not become brown due to short period of cooking.
- It is not possible to cook certain foods like chapattis, soft or hard boiled eggs, and deep fried food.
- Sometimes unwanted chemicals migrate to food from plastic cook ware or food packages.
- Short cooking time may not give a chance of blending of flavours.
- If the thickness of food is greater than 80mm the centre portion will be relatively uncooked.

(Any 4 points and each one carries 1/2 mark.)

Part C

Short Essay Type Questions.
Answer any fourquestions.
Each question carries 5 marks.

11. Physical change – change in state, shape, texture, appearance without altering the chemical composition. ***(1 mark)***

- Change in state- Freezing, melting ***(1 mark)***
- Change in size or shape- Grinding, chopping ***(1 mark)***
- Change in texture- Beating of eggwhite, kneading ***(1 mark)***
- Change in appearance- Mixing, Peeling ***(1 mark)***

12. Weather dependency***(1 mark)***
Slow cooking time-***(1 mark)***
Limited Capacity-***(1 mark)***
Requires space foe setup and operation-***(1 mark)***
Initial cost-***(1 mark)***

13 **Conduction : 2 marks**

- Heat transfer through direct contact between molecules, like when a pan heats up and transfers heat to food touching its surface- ***(1 mark)***
- This is the transfer of heat through a solid object by contact***(1 mark)***

Convection: 2 Marks

- Convection involves the transfer of heat in liquid and gases -***(1 mark)***
- . Heat transfer by the movement of a heated fluid or liquid / air, carrying heat to the food- ***(1 mark)***

Radiation: ***(1 mark)***

Radiation involves the transfer of heat by electromagnetic waves such as infra-red and microwaves.

14. **Boiling-** ***5 Marks***

- Cooking food by immersing them in water at 100 degree Celsius and maintain at that temperature till the food become tender. ***-(1 mark)***
- Water receives heat by conduction through the sides of the utensil. and passes on the heat by convection currents. ***-2 marks.***
- Boiling point of water may alter at high altitude and in the presence of electrolytes**.** ***(1 mark)***
- Foods generally cooked by boiling are rice, egg, dhal,potato,meat, sago and beet root.- ***(1 mark)***

15.Benefits : Minimizing food waste, enhancing inventory control, increasing profitability, efficient kitchen workflow, cost control, portion control

(any four points carrying ½ mark each ie, 2 marks)

Importance: Maintaining brand reputation, accurate menu pricing, improving customer satisfaction, reducing errors in food preparation, food safety standards.

(any four points carrying ½ mark each ie, 2 marks)

16. components: Recipe title, recipe category, ingredients, weight/volume of each ingredient, preparation instructions, cooking temperature & time, serving size, recipe yield, equipment & utensils to be used. ***(2 marks)***

process of standardisstion: Recipe verification, product evaluation, Quantity Adjustment. ***(2 marks)***

D

Essay Type Questions.
Answer any twoquestions.
Each question carries 10 marks.

17.1)cleaning,2) peeling and sringing,3) cutting and grating 4)Soaking 5)blanching 6)processing 7) Germination 8) fermentation 9) Grinding 10)Drying 11)filering 12) roasting.

(Explain any 5 and each one carries 2 marks)

18.**Sprouting-**
Advantages-***3 Marks***
Fermentation:
Definition ***(1 mark)***
Types -***2 mark***
Advatages-***2 mark***
Roasting
Advantages-***2 Marks***

19. Enzymatic changes- due to enzymatic reactions ***(1 mark)***

Enzymatic changes : Enzymatic browning, enzymatic fermentation, enzymatic softening & ripening, lipolysis, proteolysis.

(explain any four with definition & example carrying 1 mark each ie, 4 marks)

non-enzymatic changes- due to chemical changes without enzymes(1 mark)

Non enzymatic changes are: Maillard reaction, Caramelization, Rancidity, *Hydrolysis*
(explain any four with definition & example carrying 1 mark each ie, 4 marks)

20. Introduction of standardization (1 mark)

Components: Recipe title, recipe category, ingredients, weight/volume of each ingredient, preparation instructions, cooking temperature & time, serving size, recipe yield, equipment & utensils to be used***. (9 marks)***